

Components for LED Lighting Launched in the Last One Year

LED driver	Short feature	Company name	Feature	Application
TLC592x-Q1	Automotive 16-bit constant-current LED sink driver	Texas Instruments	• Qualified for automotive applications • AEC-Q100 Qualified with following results: 1. Device temperature grade 1: -40°C to 125°C ambient operating temperature range 2. Device HBM ESD classification level 2 • 16 constant-current output channels • Output current adjusted by an external resistor • Constant output current range: 5mA to 120mA • Constant output current invariant to load voltage change • Open-load, shorted-load and over-temperature detection • 256-step programmable global current gain • Excellent output current accuracy: 1. Between channels: < ±6% (maximum), 10mA to 50mA 2. Between ICs: < ±6% (maximum), 10mA to 50mA • 30MHz clock frequency • Schmitt trigger input • 3.3V or 5V supply voltage • Thermal shutdown for over-temperature protection • ESD performance: 2kV HBM	• General LED lighting applications • LED display systems • LED signage • Automotive LED lighting • White goods
ADP8140	4-channel high-current LED driver with adaptable power control	Analog Devices	• Highly-integrated feature set for a high-brightness LED driver solution with minimal external components • Four current sink channels with an adjustable current from 125µA to 500mA • Analogue and PWM dimming inputs • Analogue and PWM LED current outputs • Two per cent (maximum) matching between LED channels • Five per cent (maximum) LED current accuracy • Operates from VIN of 3V to 30V; higher voltages easily accomplished with an external Zener diode • Operates with LED anode supply voltages up to 100V DC • Feedback output controls external power source for optimal efficiency and safety • Multiple ADP8140 IC operation in parallel to control one power supply • Integrated error amplifier for secondary side control of isolated power supplies • Easy connection of a temperature thermistor or light sensor • Provides robust protection of the entire system 1. Power supply over-voltage protection 2. LED over-temperature protection 3. LED short-circuit protection 4. LED open-circuit protection 5. IC over-temperature protection 6. Shorted ISET protection 7. Open ISET and EN protection • Standby mode for low current consumption • Fault indicator output • Small, thermally-enhanced LFCSP package (4mm×4 mm)	• High brightness LED lighting • Large format LED backlighting
ADP8866	Charge pump-driven 9-channel LED driver with automated LED lighting effects	Analog Devices	• Charge pump with automatic gain selection of 1×, 1.5× and 2× for maximum efficiency • 92 per cent peak efficiency • Nine independent and programmable LED drivers • Each driver is capable of 25mA (full-scale) • Each driver has 7-bits (128 levels) of non-linear current settings • Standby mode for < 1µA current consumption • 16 programmable fade-in and fade-out times (0.0 sec to 1.75 sec) with choice of square or cubic rates • Automated and customisable LED blinking • Unique heartbeat mode for programmable double pulse lighting effects on four channels (D6 to D9) • PWM input for implementing content adjustable brightness control (cABC) • I2C compatible interface for all programming • Dedicated reset pin and built-in power on reset (POR) • Short-circuit, over-voltage and over-temperature protection • Internal soft start to limit inrush currents • Input to output isolation during faults or shutdown • Operates down to VIN=2.5V, with under-voltage lockout (UVLO) at 1.9 • Small lead frame chip scale package (LFCSP)	• Mobile display backlighting • Mobile phone keypad backlighting • LED indication and status lights • Automated LED blinking

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